

PolyTuf™ 1229

Introduction

PolyTuf 1229 is a micronized composite wax based on low density polyethylene (LDPE) that has been modified with two different types of high-performance ceramics. The ceramic components significantly improve abrasion resistance without the use of PTFE.

Physical Property:	PolyTuf 1229	Polyfluo 190	Polyfluo 900
Wax type	Ceramic modified polyethylene	Polyethylene/PTFE	Ceramic modified polyethylene/PTFE
Melting point, °C	110 - 113	121 - 132	121 - 132
Density @ 25 °C (g/cc)	0.97	0.98	1.02
Mean particle size (µm)	9.0 – 12.0	9.0 -12.0	9.0 – 12.0
Maximum particle size (µm)	31.11	31.11	31.11

Objective

To evaluate the performance of PolyTuf 1229 in a hard urethane formulation.

Test Methods

- ASTM D4060 Standard Test Method for Abrasion Resistance
- ASTM D3363 Standard Test Method for Film Hardness by Pencil Test (Scratch)

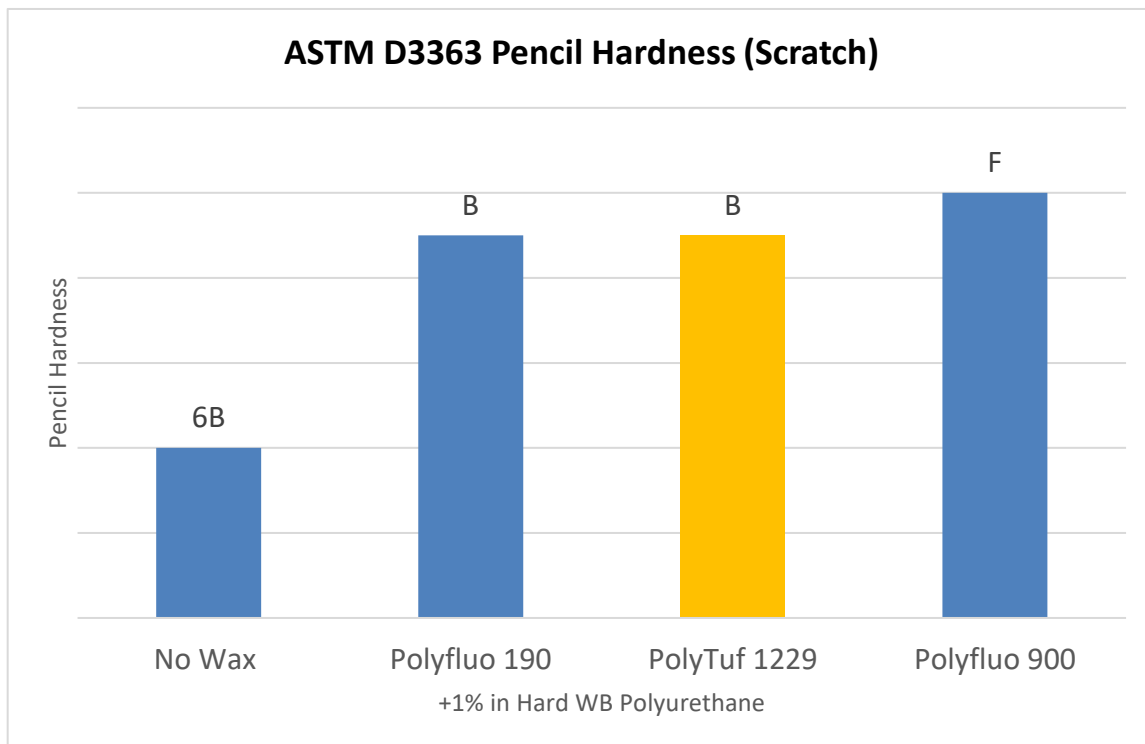
Sample Preparation

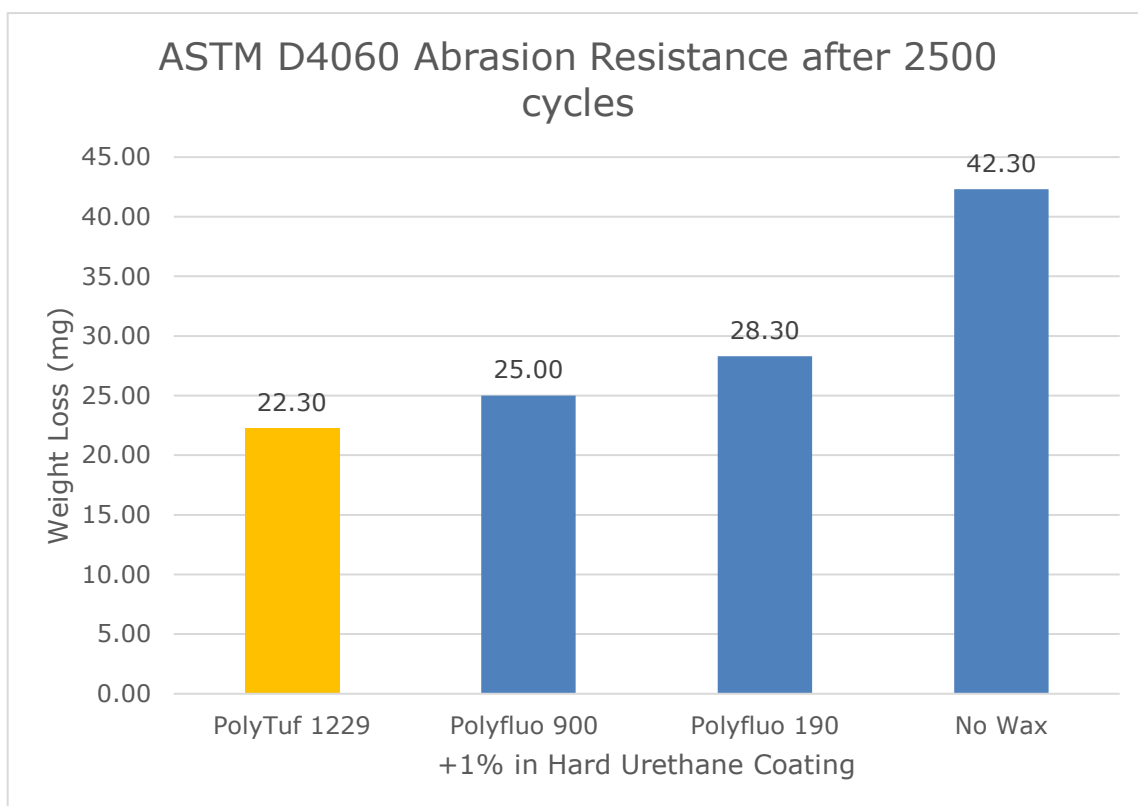
- The coatings for the Taber abrasion test were drawn down with a #60 wire wound rod to produce approximately 2.4 mils (60 µm) DFT coatings.
- The coatings for scratch resistance testing were drawn down with a #28 wire wound rod to produce approximately 1.0 mil (25 µm) DFT coatings.
- All coatings are allowed to cure for 7 days at room temperature at ~50% humidity.

Formulation:

Hard Urethane Formula:	Weight %:
UROTUF L63-MPW-38 (Reichhold)	97.5
Diethylene glycol monoethyl ether	2.0
BYK-345	0.2
TEGO Foamex 805 N	0.3
Total:	100.0

Results





Conclusions

Scratch resistance

- PolyTuf 1229 has similar scratch properties to Polyfluo 190, and almost as good as Polyfluo 900
- PolyTuf 1229 is a good PTFE-free option for either of these products

Abrasion resistance

- PolyTuf 1229 has the best abrasion data of all MPI fine powder products, with a 22.3 mg weight loss after 2,500 cycles.
- The abrasion resistance performance of PolyTuf 1229 is a 10.8% improvement over Polyfluo 900.